Geometry
Unit 9
Lesson 6 Homework

Name____Answer Key______
Date_____________________
Period___________________

1. Find the volume of a rectangular prism with a width of 7.2 feet, a length of 6.9 feet, and a height of 10.5 feet.
$V = (7.2)(6.9)(10.5) = 521.64 \text{ ft}^3$

2. Find the volume of a pyramid whose height is 10.5 feet and whose base is a rectangle with dimensions of 7.2 feet by 6.9 feet.
$V = \frac{1}{3}(7.2)(6.9)(10.5) = 173.88 \text{ ft}^3$

3. What is the ratio of the answer from Question 1 to the answer from Question 2?
$\frac{521.64}{173.88} = 3$
Ratio is $3:1$.

4. Find the volume of a sphere with a diameter of 1.74 meters. Use 3.14 for π. Round to the nearest hundredth.
$V = \frac{4}{3}\pi(0.87)^3 \approx 2.76 \text{ m}^3$

5. Find the volume of a hemisphere with a diameter of 1.74 meters. Use 3.14 for π. Round to the nearest hundredth.
$V = \frac{2}{3}\pi(0.87)^3 \approx 1.38 \text{ m}^3$

6. What is the relationship between the answers from Questions 4 and 5?
The hemisphere has half the volume of the sphere.

7. Determine the volume of a cone with a radius of 16.2 inches, a height of 5.75 inches, and a slant height of 9.9 inches in terms of π.

$V = \frac{1}{3}\pi(16.2)^2(5.75)$

$V = 503.01\pi$ in^3

8. Find the volume of a cylinder with a radius of 16.2 inches and a height of 5.75 inches. Round your answer to the nearest thousandth.

$V = \pi(16.2)^2(5.75) \approx 4740.758$ in^3 (when using the pi button)

9. A composite figure consists of a cylinder and a square pyramid. The height of the cylinder is 58.1 centimeters. The height of the pyramid is 23 centimeters. The base of the square pyramid has a width of 19 centimeters. The base of the pyramid sits on one base of the cylinder with the width of the square base equal to the diameter of the cylinder. Find the volume of the composite figure. Round to the nearest hundredth.

Cylinder: $\pi(9.5)^2(58.1) = 16473.02$ (when using the pi button)

Pyramid: $\frac{1}{3}(19)(19)(23) = 2767.67$

$16473.02 + 2767.67 = 19240.69$ cm^3

10. A composite figure consists of a cylinder and a hemisphere. The height of the cylinder is 32.5 inches. The base of the hemisphere sits on one base of the cylinder, both having a diameter of 14 inches. Find the volume of the composite figure. Use 3.14 for π. Round to the nearest hundredth.

Cylinder: $V = \pi(7)^2(32.5) = 5000.45$

Hemisphere: $V = \frac{2}{3}\pi(7)^3 = 718.01$

$5000.45 + 718.01 = 5718.46$ in^3